Global Electronics Retailer Data Analysis

1. Project Overview

The Global Electronics Retailer project focuses on analyzing retail data to understand sales performance, product trends, and customer purchasing behavior. Excel is used for data cleaning and preliminary analysis, while Power BI is used to create interactive dashboards and visualizations. The project transforms raw data into meaningful business insights to support effective decision-making.

Project Objective

The objective is to analyze the Global Electronics Retailer dataset using Excel and Power BI to identify sales trends and product performance. It involves data cleaning, exploratory data analysis, and creating charts, PivotTables, and interactive dashboards. The project aims to generate actionable business insights to improve retail performance and support better decision-making.

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2. Data Sources

- **Source Description and Timeline:** Maven Analytics
- **Domain**: Sales Analytics

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3. Problem Statement

The Global Electronics Retailer manages large volumes of data related to sales, products, customers, and stores. However, analyzing these datasets manually makes it difficult to identify sales trends, understand product performance, evaluate customer distribution, and monitor store operations effectively. The lack of an integrated analytical approach limits the ability to identify business opportunities and make informed decisions. Therefore, this project aims to clean, analyze, and visualize the retail datasets using Microsoft Excel and Power BI to identify meaningful business insights, improve business performance analysis, and support data-driven decision-making.

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4. Attribute (Column /Features) Details

Attribute Name	Data Type	Description
CustomerKey	Integer	Unique identifier for each customer.
Gender	Text	Gender of the customer.
Name	Text	Full name of the customer.

City	Text	City where the customer lives.
Code & State	Text	State code and state name of the customer.
Country	Text	Country where the customer lives or the store is located.
Continent	Text	Continent where the customer lives.
Birthday	Date	Date of birth of the customer.
Date	Date	Date on which the exchange rate was recorded.
Currency	Text	Currency code for the exchange rate.
Exchange	Decimal	Exchange rate for the specified currency and date.
ProductKey	Integer	Unique identifier for each product.
Product Name	Text	Name of the product.
Brand	Text	Brand associated with the product.
Color	Text	Color of the product.
Manufacture Price	Decimal	Cost of manufacturing the product.
Selling Price	Decimal	Price at which the product is sold.
SubcategoryKey	Integer	Unique identifier for the product subcategory.
Subcategory	Text	Subcategory to which the product belongs.
CategoryKey	Integer	Unique identifier for the product category.
Category	Text	Category to which the product belongs.
Order Number	Integer	Unique identifier for the sales order.
Line Item	Integer	Identifies an individual item within an order.
Order Date	Date	Date on which the order was placed.
Delivery Date	Date/Text	Delivery date of the order or a status such as "Not Delivered".
StoreKey	Integer	Unique identifier for each store.
Quantity	Integer	Number of units purchased.
Currency Code	Text	Currency code used for the sales transaction.
State	Text	State where the store is located.
Square Meters	Integer	Physical area of the store in square meters.
Store Size Categories	Text	Category representing the store's size.
Open Date	Date	Date on which the store opened.

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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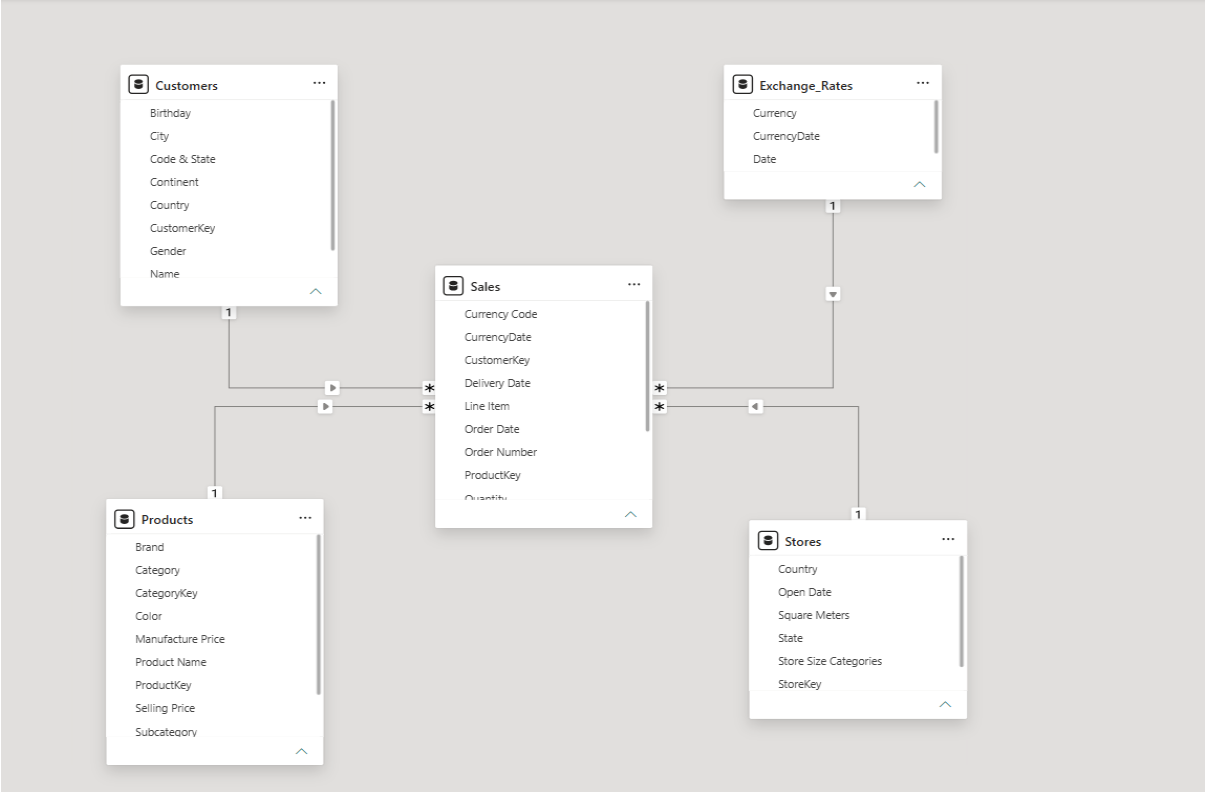
6. Data Pre-Processing (Excel / Power Query)

- Removed unwanted columns for analysis purpose
- Format Table using format as table
- Changed Data types Respectively for each column
- Used Removing Duplicates in event Id column – No duplicates found in my data
- Splits Event Date and Admitted Date columns used Text to Columns by delimiter to split date and time
- Did Descriptive Statistics analysis using toolpak
- IF(ISBLANK(A2),"Unknown",A2) – Used to handle missing values in text columns of the Customers, Products, Sales, Stores, and Exchange Rates datasets.
- PROPER(TRIM(CLEAN(A2))) – Used to remove extra spaces, eliminate non-printable characters, and standardize text formatting across the datasets.
- TEXT(A2,"dd-mm-yyyy") – Used to standardize date formatting in the Sales and Exchange Rates datasets, where applicable.
- Used Pivot Tables in Excel to summarize, organize, and analyze data from the cleaned datasets. Pivot Tables helped calculate totals, averages, and counts, making it easier to identify patterns and compare information across different categories.

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- **Calculated Columns & DAX Measures:**

- **Total Sales**

  Total Sales = SUM('Sales Fact'[Amount])
- **Total Quantity**

  Total Quantity = SUM('Sales Fact'[Quantity])
- **Total Orders**

  Total Orders = DISTINCTCOUNT('Sales Fact'[Order ID])
- **Total Customers**

  Total Customers = DISTINCTCOUNT('Customer DIM'[CustomerKey])
- **Total Stores**

  Total Stores = DISTINCTCOUNT('Store DIM'[StoreKey])

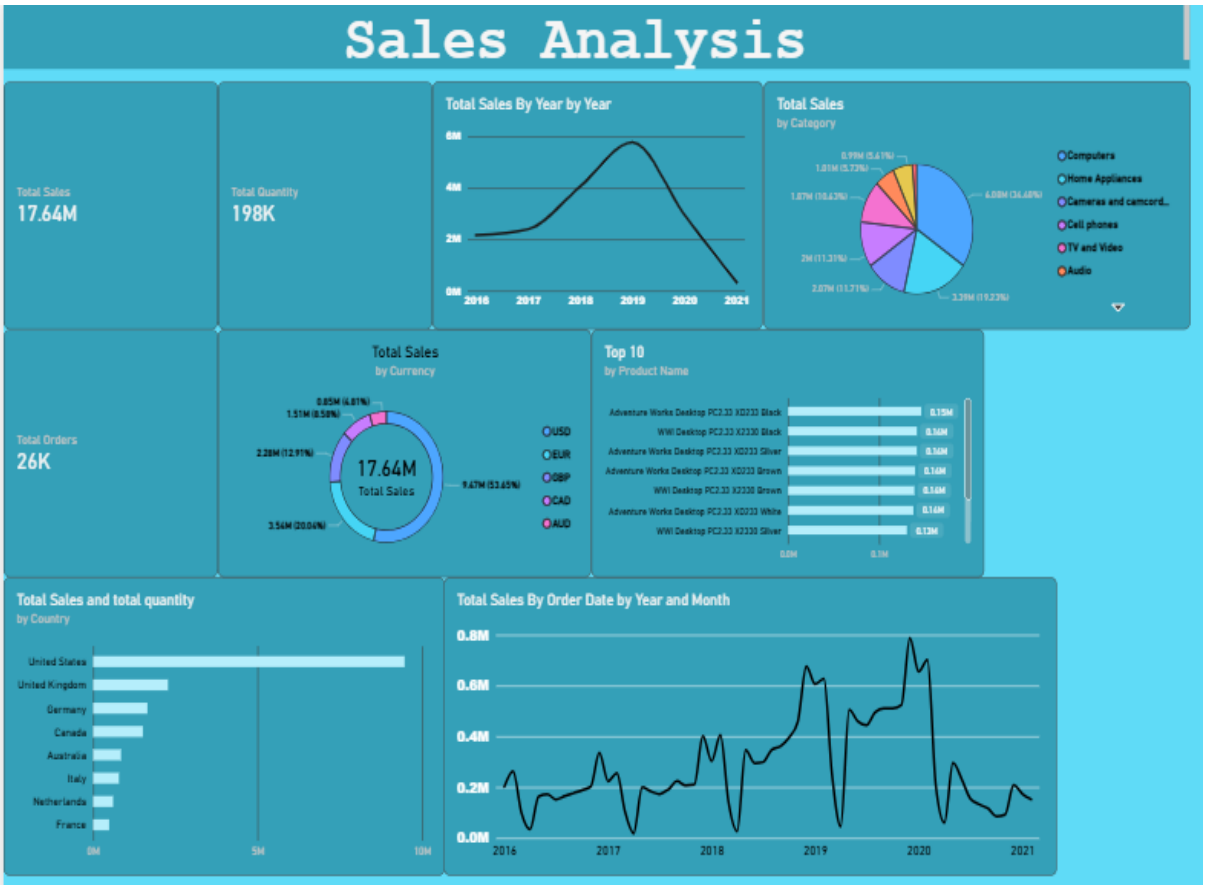
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8. Analysis and Visualizations (Power BI)

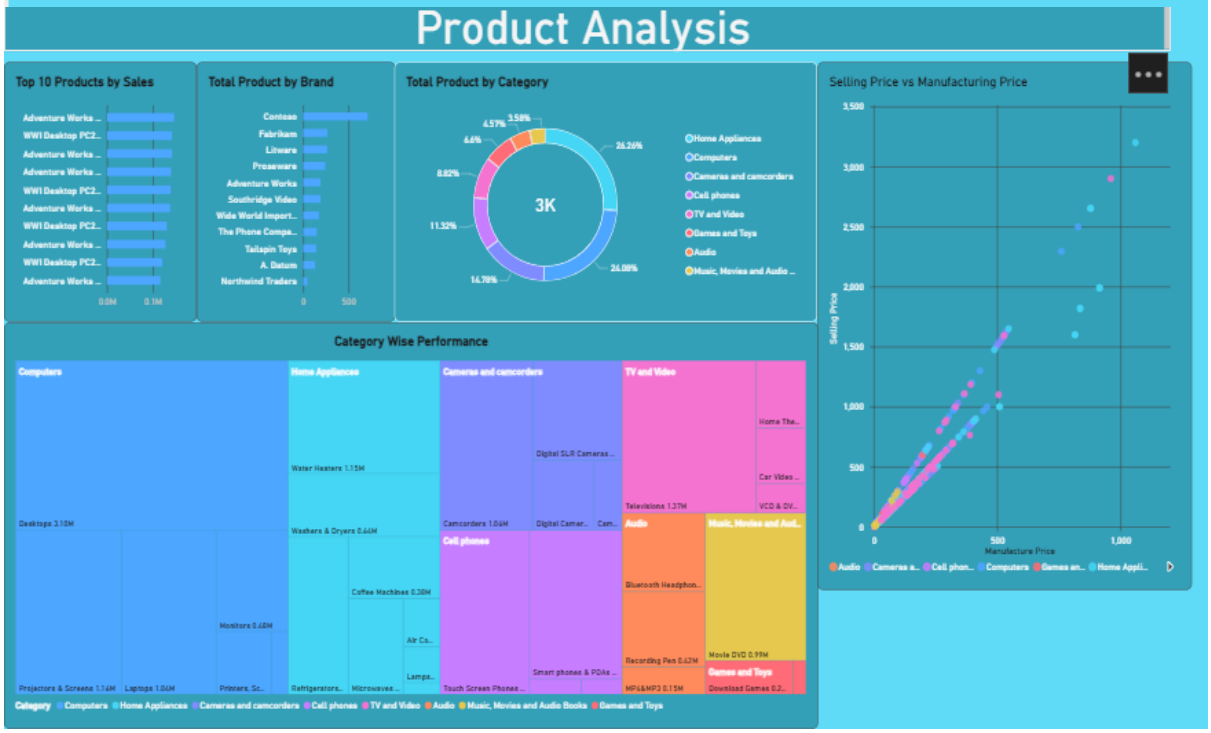
Dashboard Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights. **(List Out)**
- Make your Report interactive Layout using **Drill-down**, **Filters and slicers**

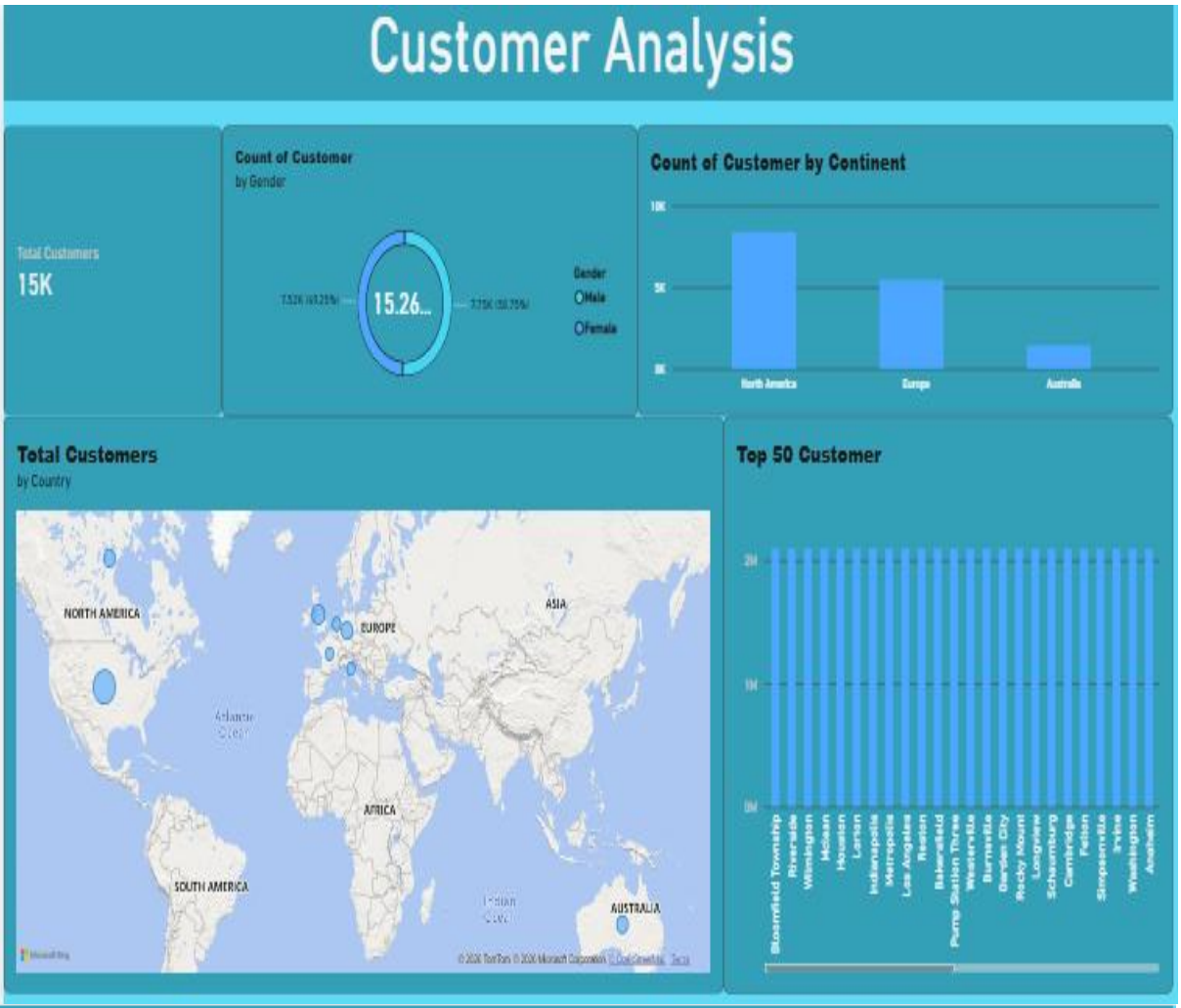
Sales Analysis



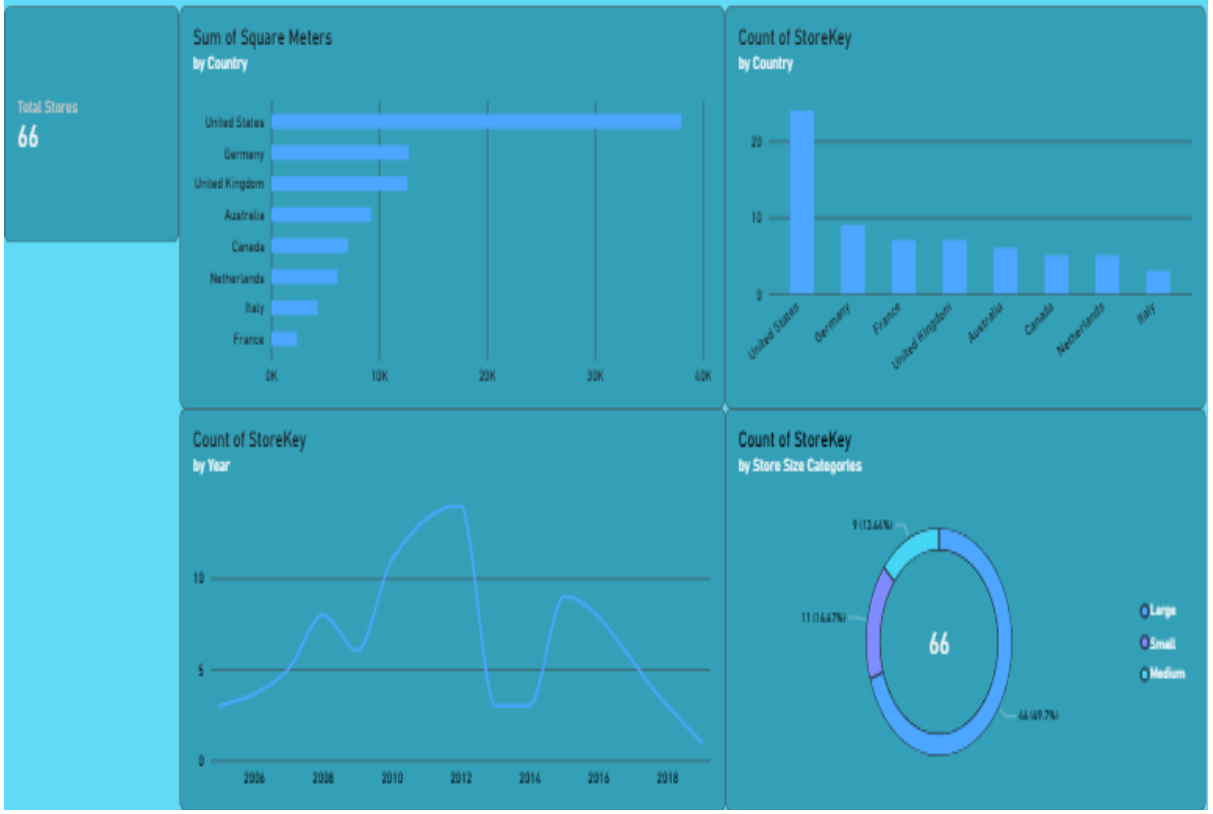
Product Analysis



Customer Analysis



Store Analysis



9. Insights & Conclusions

- **Key Findings:** The analysis of the Global Electronics Retailer dataset revealed important trends in sales, products, customers, and stores. The business generated approximately 17.64 million in total sales, with around 198K units sold and 26K orders. The United States emerged as the leading sales market. Product analysis highlighted differences in sales across categories and brands. Customer analysis revealed geographical and gender distribution patterns, while store analysis identified variations in store size and distribution across countries. These findings help understand business performance and identify potential areas for improvement.
- **Provide the analysis insights:**
  - **Descriptive :**

    Descriptive analytics focuses on analyzing historical data to understand business performance. In this project, Power BI was used to analyze total sales, quantity sold, orders, product categories, customer demographics, and store distribution to identify historical trends and patterns.
  - **Diagnosis :**

    Diagnostic analytics helps investigate factors influencing business performance. This project examines sales variations across countries, product categories, brands, and customer segments to identify performance differences and potential areas for improvement.
  - **Predictive :**

    Predictive analytics utilizes historical data to estimate future business outcomes. The sales trends and product performance analyzed in this project can support future sales forecasting, demand prediction, and customer purchasing behavior analysis.
  - **Prescriptive :**

    Prescriptive analytics provides recommendations to improve business performance. Based on the project findings, potential strategies include inventory optimization, targeted marketing campaigns, product promotion, and expansion into high-performing markets.

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10. Conclusions

Conclusion

The Global Electronics Retailer project successfully analyzed sales, product, customer, and store data using Excel and Power BI. Through data cleaning, DAX calculations, and interactive dashboards, key business insights were identified, including sales performance, product trends, customer distribution, and store operations. The project demonstrates the application of

descriptive and diagnostic analytics to understand business performance. The findings also support future predictive sales forecasting and prescriptive business recommendations for inventory optimization, marketing strategies, and improved decision-making. Overall, this project enhances business understanding and supports data-driven decision-making.